



MediaMix: Multimedia Retrieval in Mixed Reality

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Abstract. Extended reality (XR), which encompasses the concepts of augmented (AR), mixed (MR), and virtual reality (VR), enables unique user interactions. Focusing on mixed reality is essential to enabling interaction between the virtual and the real world. One application in which MR can have a significant impact is multimedia retrieval. Moving away from desktop user interfaces towards MR creates new, immersive, and more natural ways for users to interact with the retrieval engine, which can enhance the user experience. In an MR world, there are many new ways of creating queries and interacting with results, which are much more intuitive and user-friendly than in a traditional 2D setting.

By introducing MediaMix, a multimedia retrieval system in mixed reality, we introduce a concept and system for user interactions regarding query generation and result interaction in an MR environment. MediaMix will participate in the Video Browser Showdown (VBS) 2025 for the first time. The system utilises the newest MR hardware, the Apple Vision Pro, to facilitate easy user input for query generation and investigates new result presentation and interaction utilising eye tracking and gesture recognition. For the retrieval backend, vitrivr-engine, connected to Postgres, is employed to provide advanced retrieval functionality.

Keywords: Video Browser Showdown · Interactive Video Retrieval · Mixed Reality

1 Introduction

Mixed reality (MR) [18] has emerged as a promising paradigm that seamlessly integrates virtual and real-world elements. By enabling users to interact with both physical and digital objects in a single immersive environment, MR offers distinctive opportunities for innovation in various domains like education, healthcare, and the gaming industry. Multimedia retrieval, and in particular video retrieval, seamlessly supplements this list.

The specific domain of video retrieval has seen significant advancements in recent years, driven by increased online video content and the growing demand

for efficient searching and browsing mechanisms in everyday systems. Traditional video retrieval systems rely on query-based 2D desktop interfaces, which can be cumbersome and ineffective in handling complex queries or large datasets. As a result, explorations have been made into alternative approaches to improve user engagement and interaction with video collections.

Compared to most systems in video retrieval, our research introduces mixed reality as an interaction pattern and as a platform for multimedia retrieval, including video browsing. By leveraging MR's interactive capabilities, we aim to enhance the search, especially the browsing experience, creating more immersive and dynamic search experiences. Our goal is to empower users to navigate extensive video collections effectively. This paper presents MediaMix, a new innovative MR-based retrieval system that has the potential to enhance user interaction through advanced gesture recognition and eye-tracking, running on the Apple Vision Pro.

This work will present our approach by showcasing the concept and prototype MediaMix at the VBS 2025 competition. By participating in VBS, we hope to provide a compelling benchmark for existing video retrieval systems and highlight the potential benefits of incorporating MR into multimedia search applications.

2 Mixed Reality and Multimedia Retrieval

In this section, we first provide an overview of mixed reality followed by multimedia retrieval, focusing on multimedia retrieval in extended reality.

2.1 Mixed Reality

Mixed Reality [13, 18] encompasses a spectrum of experiences that fall between Virtual Reality (VR) and Augmented Reality (AR). While VR immerses users in a wholly digital environment, AR overlays digital content onto the physical world. By blurring the boundaries between physical and virtual domains, MR opens up exciting new possibilities for interactive and immersive experiences. The potential applications for MR are extensive, spanning industries such as entertainment [10, 23], education [17, 26], and healthcare [5, 8]. For example, MR can create unforgettable gaming experiences that seamlessly blend virtual objects with the real world or overlay educational content onto real-world objects to facilitate interactive learning. In healthcare, MR is used for surgical planning and training, while the industry relies on it for design visualisation and remote collaboration. Mixed reality hardware encompasses a range of devices, from headsets like the Apple Vision Pro¹, the Microsoft HoloLens² or the Meta Quest Pro³ to handheld devices such as smartphones and tablets. These hardware components are designed to seamlessly blend digital and physical worlds, enabling immersive and interactive MR experiences.

¹ <https://www.apple.com/apple-vision-pro>.

² <https://www.microsoft.com/en/hololens>.

³ <https://www.meta.com/en/quest/quest-pro>.

2.2 Multimedia Retrieval

Multimedia retrieval involves searching for and retrieving multimedia content from a dataset based on user queries, including images, videos, audio, and text. It is critical in various applications, such as content-based image retrieval and video recommendation. Multimedia retrieval presents a unique challenge that differentiates it from structured data retrieval. It involves a dataset of N documents ($D_0, \dots, D_{N-1}$) and a query Q , with the primary goal of producing a (ranked) list of documents D_j ($0 < j < M$, where $M \leq N$) that match the query's criteria. The ranking is based on relevance, indicating a document's similarity concerning the query. The query can use multiple features, each representing a distinct aspect of the multimedia file. Depending on the context, multimedia retrieval serves various purposes. Sometimes, the objective is to uncover all relevant information, as in AVS (ad-hoc video search) tasks at VBS. Each AVS task is described in text, and a unique correct answer does not exist. Numerous target segments must be found quickly to build the results. In contrast, there are situations where finding a single suitable answer is sufficient, with little concern for the existence of multiple alternatives. This scenario is captured in KIS (known-item search) tasks at VBS, where a segment of a video clip or textual description is given. Only one target segment is the correct result. Additionally, multimedia retrieval and, therefore also, video retrieval can be utilised as well to find an answer to a question (Q/A tasks at VBS).

vitrivr-VR [19, 20] represents an innovative leap into multimedia retrieval, as the system's user interface is entirely in VR. The system is connected to Cineast, the first backend of vitrivr, a 2D multimedia retrieval tool [14]. The system is focused on text input in VR, as there is no possibility of using a standard physical keyboard in VR [22]. Furthermore, much effort is put into result browsing creating the multimedia drawer which enhances the user-experience [21].

Virtual Reality Multimedia Analytics (ViRMA) [4] is another VR system exploiting sophisticated projections of multimedia objects for visual analytics in virtual spaces. It is based on a multi-dimensional multimedia model (M^3) [9] from which objects are projected onto a three-dimensional space in VR. Traditional Data Warehousing operations like slicing & dicing, pivoting, or drill-down and roll-up have been adjusted to the multimedia context.

3 MediaMix – System Overview

The MediaMix system uses a retrieval system called vitrivr-engine as its backend [6]. vitrivr-engine is the latest retrieval backend, part of the vitrivr stack, and it replaced Cineast [16] as of VBS 2023. The new backend is much more modular and flexible. Thanks to its modularity, PostgreSQL [24] with the pgvector⁴ extension is used as a database for VBS 2025. In recent years, systems based on the vitrivr stack have still used Cottontail DB [7]. With this change, we aim to achieve better performance and faster query results for retrieving data from the

⁴ <https://github.com/pgvector/pgvector>.

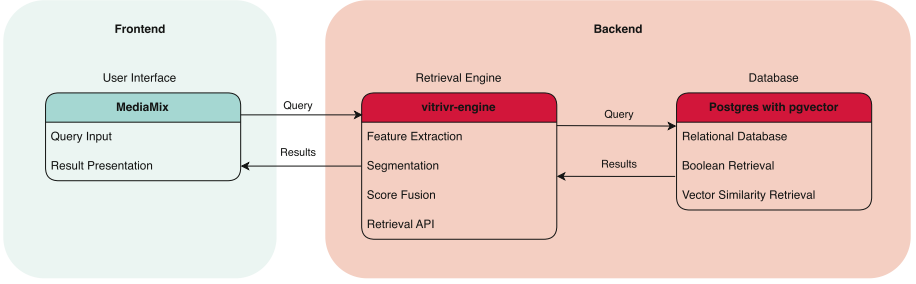


Fig. 1. Architecture overview of the MediaMix stack

two datasets (V3C [15] and marine video kit [25]) used at VBS 2025. Figure 1 illustrates the three-tier architecture.

In the remainder of this section, we will first provide information about query formulation before providing an overview of the result presentation and the possibility of user interaction.

3.1 Query Input and Features

There are multiple options for creating queries in MR. On the one hand, traditional query methods from the 2D world can be utilised, such as text descriptions, OCR, ASR, tags, examples, motion, and audio. However, capturing live footage of the natural world to create queries offers exciting possibilities [1]. This enables MediaMix users to interact with the real world directly. For VBS 2025, we restrict the possible query formats to text and similarity queries. Live queries are not utilised to adhere to VBS guidelines, and no tasks are directly scanned.

Text queries can be created in two different ways. Users can input text through a physical keyboard, which is then used for the semantic search. We intend to incorporate CLIP as a feature [3, 12]. Alternatively, users can verbally speak queries, which are then converted using speech-to-text. For transcription, we use Apple’s speech-to-text transcription service.

For the similarity search, we use individual frames from videos. These frames are indexed and utilised for a similarity search that can be performed using CLIP or DINO [11].

3.2 Result Presentation and User Interaction

In MediaMix, search results are visually represented on a dynamic, interactive globe. This globe-based display draws inspiration from navigu.net, where the results are mapped across the surface of a sphere [2]. Additionally, contextually relevant results appear as users zoom out, enriching the exploration experience. Conversely, zooming in reduces the number of displayed results, enabling a more focused view of critical information.

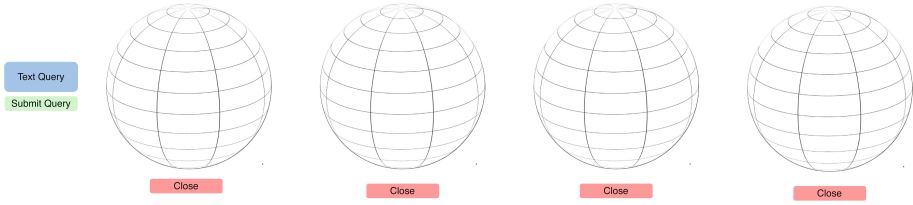


Fig. 2. The graphic shows the query interface and the results of four different queries. Each query’s results are displayed on a separate globe, which is displayed until the user closes it again.

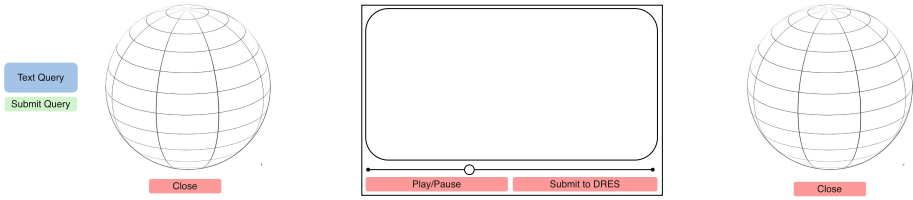


Fig. 3. The input for the query is on the left side, along with a button to submit it. The results are presented on a globe next to it, where the pictures of the keyframes of the retrieved segments are used. If a result is selected, a video player and another globe are opened, which presents all segments of this specific video.

Each search query generates its dedicated globe, organising results in a spatial, visually intuitive manner. When new queries are made, previously created globes remain visible in the environment, as seen in Fig. 2. This ensures that users always see earlier results, facilitating a continuous exploration process. Rather than overwriting past searches, the system allows users to compare and contrast multiple search queries simultaneously, fostering deeper insight and discovery. When a user selects a result, a 2D plane opens with a video player (Fig. 3). In this interface, users can play and navigate through the video. A globe displaying the individual frames will also appear for selected videos, providing a quick look and overview of the multimedia object.

Users navigate the system using intuitive gestures and gaze-based controls, providing a seamless, immersive experience. They can freely rotate and reposition the globe, offering complete control over their exploration. The sphere’s interaction model allows users to explore, manipulate, and compare search results in a visually engaging, multi-dimensional mixed reality space.

4 Conclusion

In this paper, we introduced and provided an overview of our new mixed reality multimedia retrieval system MediaMix. We highlighted some fundamentals of MR and covered the novel query and result user interactions with which we plan to participate at VBS 2025.

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